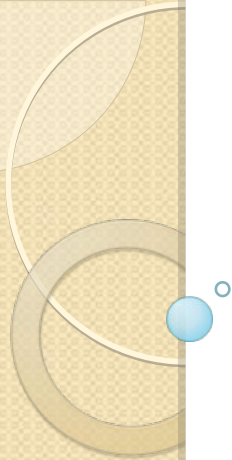




Toxicology

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1. Introduction to Toxicology

Upon completion of this session, the you will be able to:

1. Define toxicology and commonly associated terms
2. Differentiate the sub-disciplines of toxicology
3. Describe the field of toxicology
4. Understand the applications of toxicology

1.1 What is Toxicology? (defn)

- Traditional definition of **toxicology** is "***the science of poisons.***"
- ***What is a poison?***
- ***What substance is there that is not a poison?***

All substances are poisons;
there is none that is not a poison. The right
dose differentiates a poison and a remedy.

Paracelsus (1493-1541)

What is toxicology contd.

- A more descriptive definition of toxicology is *"the study of the adverse effects of chemicals or physical agents on living organisms"*.
- What is adverse effects?
 - any change from an organism's normal state
 - may occur in many forms, ranging from immediate death to subtle changes not realized until months or years later.
 - may occur at various levels within the body, such as an organ, a type of cell, or a specific biochemical

Paracelsus

**“All substances are poisons;
there is none which is not a poison.
The right dose differentiates a poison
from a remedy.”**

Paracelsus (1493-1541)

An Individual View

“The sensitivity of the individual differentiates a poison from a remedy. The fundamental principle of toxicology is the individual’s response to a dose.”

S. G. Gilbert (1997)

1.2 Brief history of toxicology

- early cave dwellers recognized poisonous plants and animals and used their extracts for hunting or in warfare or for assassination.
- By 1500 BC, written recordings indicated that hemlock, opium, arrow poisons, and certain metals such as lead, copper and antimony were used to poison enemies or for state executions.

History of toxicology contd.

- With time, poisons became widely used and with great sophistication.
 - Notable poisoning victims include Socrates, Cleopatra, and Claudius.
- By the time of the Renaissance and Age of Enlightenment, certain concepts fundamental to toxicology began to take shape.
 - Noteworthy in this regard were the studies of Paracelsus (~1500AD) and Orfila (~1800 AD).

History of toxicology contd.

- **Paracelsus** determined that specific chemicals were actually responsible for the toxicity of a plant or animal poison.
- He also documented that the body's response to those chemicals depended on the dose received.
- His studies revealed that small doses of a substance might be harmless or beneficial whereas larger doses could be toxic.
 - This is now known as the dose-response relationship, a major concept of toxicology.

History of toxicology contd.

- **Orfila**, a Spanish physician, is often referred to as the **founder of toxicology**.
- It was Orfila who first prepared a systematic correlation between the chemical and biological properties of poisons of the time.
- He demonstrated effects of poisons on specific organs by analyzing autopsy materials for poisons and their associated tissue damage.

History of toxicology contd.

The 20th century

- is marked by an advanced level of understanding of toxicology.
- DNA (*the molecule of life*) and various biochemicals that maintain body functions were discovered.
- Our level of knowledge of toxic effects on organs and cells is now being revealed at the molecular level.
- It is recognized that virtually all toxic effects are caused by changes in specific cellular molecules and biochemicals.

1.3 *Basic Toxicology Terminology*

- Toxicology is an evolving medical science and so is toxicology terminology
 - Those most commonly used are discussed in the following frames.
- **Toxicology** is the study of the adverse effects of chemicals or physical agents on living organisms.
- A **toxicologist** is a scientist that determines the harmful effects of agents and the cellular, biochemical, and molecular mechanisms responsible for the effects.

Toxicology terminology contd.

- Terminology and definitions for materials that cause toxic effects are not always consistently used in the literature.
- The most common terms are **toxicant, toxin, poison, toxic agent, toxic substance, and toxic chemical.**
- Toxicant, toxin, and poison are often used interchangeably in the literature; however, there are subtle differences as indicated below:

Toxicology terminology contd.

- Toxicants
 - Substances that produce adverse biological effects of any nature
 - May be chemical or physical in nature
 - Effects may be of various types(acute, chronic etc)
- Toxins
 - Specific proteins produced by living organisms (mushroom toxin, tetanus toxin)
- Poisons
 - toxicants that cause immediate death or illness when experienced in very small amounts

Toxicology terminology contd.

A toxic agent

- is anything that can produce an adverse biological effect.
- It may be chemical, physical, or biological in form.
 - For example, toxic agents may be chemical (*such as cyanide*), physical (*such as radiation*) and biological (*such as snake venom*).
- A distinction is made for diseases due to biological organisms.
 - Those organisms that invade and multiply within the organism and produce their effects by biological activity are not classified as toxic agents. An example of this is a virus that damages cell membranes resulting in cell death.

Toxicology terminology contd.

- A **toxic substance** is simply a material which has toxic properties.
- It may be a discrete toxic chemical or a mixture of toxic chemicals. For example, lead chromate, asbestos, and gasoline are all toxic substances.
 - Lead chromate is a discrete **toxic chemical**.
 - Asbestos is a **toxic material** that does not consist of an exact chemical composition but a variety of fibers and minerals.
 - Gasoline is also a **toxic substance** rather than a toxic chemical in that it contains a mixture of many chemicals.

1.4 Areas, Scope and Applications of Toxicology

- Mechanistic toxicologist
 - Identifies the cellular, biochemical, and molecular mechanisms by which chemicals exert toxic effects on living organisms
 - useful in the design and production of safer chemicals and
 - in rational therapy for chemical poisoning and treatment of disease.
 - **Toxicogenomics**
 - Identify and protect genetically susceptible individuals from harmful environmental exposures (genetic makeup)

Different areas of Toxicology (cont.)

- Descriptive toxicologist
 - Toxicity testing
 - Provides information for safety evaluation
 - Regulatory requirements
 - to evaluate the risks posed to humans and the environment by exposure to specific chemicals.
- Regulatory toxicologist
 - Deciding using data whether a drug or chemical poses a risk (low or high)
 - Standards (air, water, land)

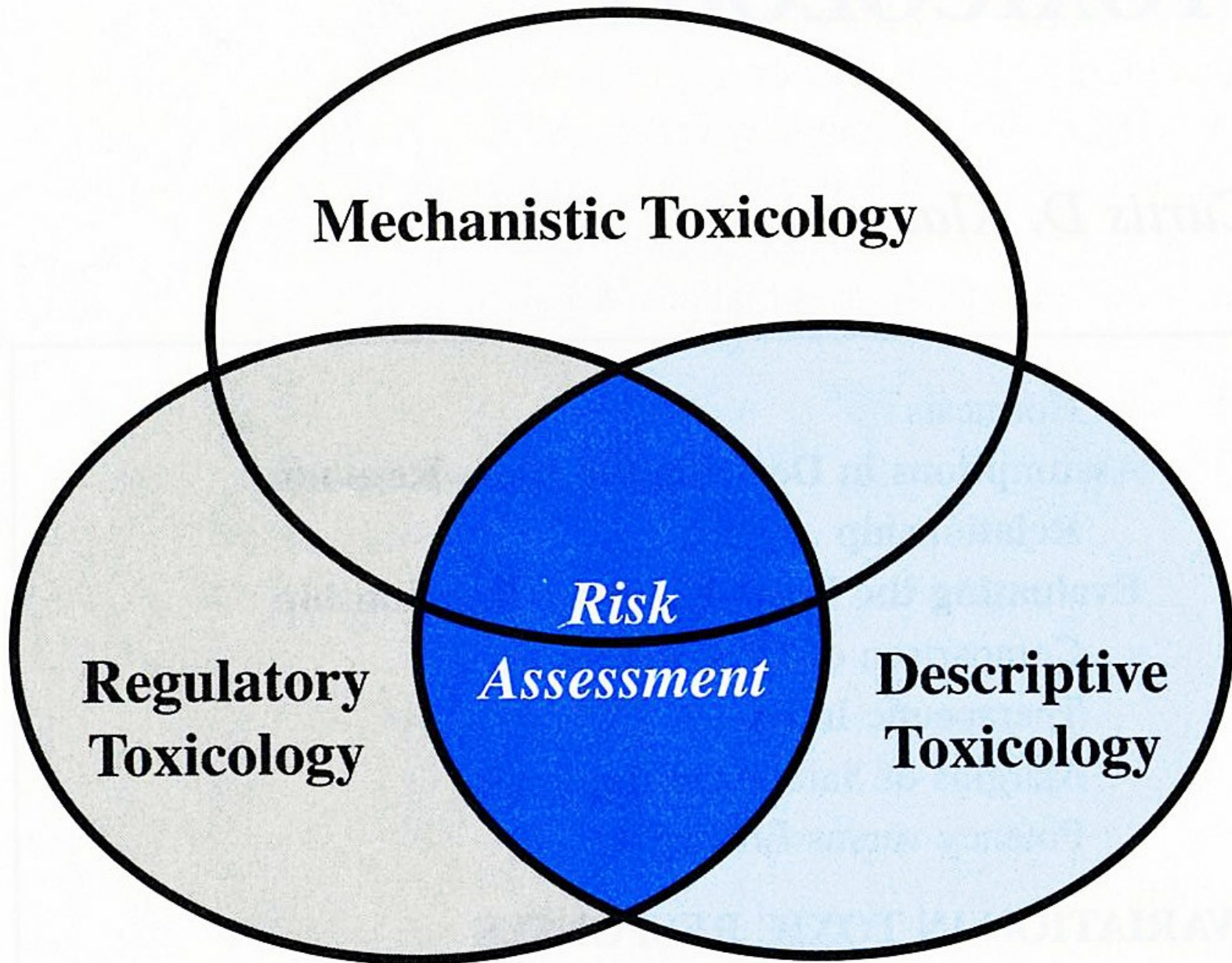


Figure 2-1. Graphical representation of the interconnections between different areas of toxicology.

Different areas continued

- Forensic toxicologist
 - Analytic chemistry and toxicology
 - Medicolegal aspects of harmful effects of chemicals on humans
- Clinical toxicologist
 - Diseases caused by or associated with toxic substances

Different areas of toxicology (cont.)

- **Environmental toxicologist**
 - Impacts of chemical pollutants in the environment on biological organisms (fish, birds, etc.)
 - **Ecotoxicology**- impacts of toxic substances on population dynamics in an ecosystem
- **Developmental toxicologist**
 - Study of adverse effects on the developing organism.
 - **Teratology**- study of defects induced during development
- **Reproductive toxicologist**- adverse effects on the male or female reproductive system

1.5 Why toxicology?: rationale

- *Humans are exposed to chemicals both inadvertently and deliberately.*
- Exposures to chemical, physical and biological agents have the potential to adversely affect human health.
- Toxicologists
 - identify environmental exposures of concern,
 - reveal mechanisms by which toxicants initiate pathological dysfunction or disease,
 - identify interventions for the prevention of adverse effects, and
 - estimate acceptable levels of exposure for the protection of the public health

Experimental Toxicology

Dose_Response_Relationship

- Paracelsus is noted for stating the concept that “THE DOSE MAKES THE POISON.”
- At extremely low doses, a given substance may be non-toxic and even beneficial (a concept known as hormesis), while at intermediate doses, it may be toxic.
- At high doses, it may be lethal.
- Determining the LD50 of a drug, compound or toxicant is an important first step in discerning its relative toxicity
- underscores the importance of understanding dose response relationships.

Dose

The amount of chemical entering the body

This is usually given as

mg of chemical/kg of body weight = mg/kg

The dose is dependent upon

- * The environmental concentration
- * The properties of the toxicant
- * The frequency of exposure
- * The length of exposure
- * The exposure pathway



Woman Dies after Water-drinking Contest: Water Intoxication???

SACRAMENTO, California—A woman who competed in a radio station's contest to see how Much water she could drink without going to the bathroom died of water intoxication, the coroner's office said Saturday (AP). Updated: 10:24 p.m. ET Jan 13, 2007

What is a Response?

The degree and spectra of responses depend upon the dose and the organism--describe exposure conditions with description of dose

- Change from normal state
 - could be on the molecular, cellular, organ, or organism level--the symptoms
- Local vs. Systemic
- Reversible vs. Irreversible
- Immediate vs. Delayed
- Graded vs. Quantal
 - degrees of the same damage vs. all or none

Measure of Response?

- **Death**

- toxic response that is simplest to observe is
- But it is a crude parameter to measure

- **Pathological lesion (Liver cell necrosis)**

~

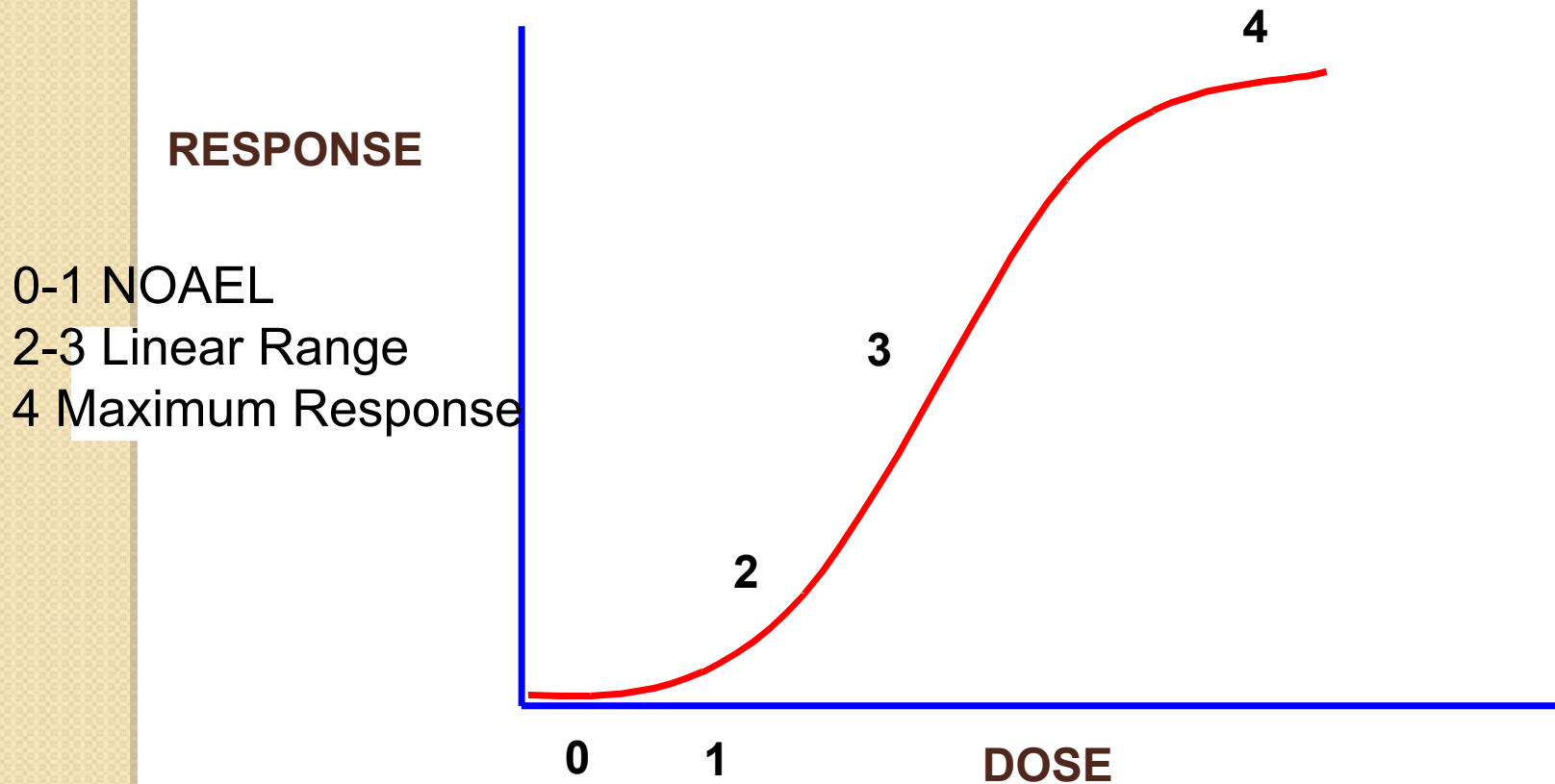
- indicator of toxic responses.

- **Biochemical, Pharmacological or Chemical change**

- More precise measure of responses

Dose-Response Relationship:

As the dose of a toxicant increases,
so does the response.



DOSE DETERMINES THE BIOLOGICAL RESPONSE

DOSE RESPONSE

-ASSUMPTIONS

- response is due to chemical administered
- the response is related to the dose
 - there is a receptor site with which the chemical interacts

DOSE RESPONSE

-ASSUMPTIONS (contd)

- the degree of response is related to the concentration at the site
- the concentration at the site is related to the dose administered
- has a quantifiable method of measuring and a precise means of expressing the toxicity

Toxicokinetics (cont.)

- Absorption
 - A process by which the toxicant gets transferred to the circulation
 - It depends on the route of exposure
 - Fastest from inhalation exposure

Toxicokinetics (cont.)

- Distribution
 - Affected by
 - PPB
 - RBF
 - Tissue affinity
 - Presence of barriers

Toxicokinetics (cont.)

- Biotransformation
 - Alteration of toxicants in the body
 - May lead to more toxic metabolites (parathion to parathoxon)
 - Toxicants taken orally subjected to 1st pass effect
 - Variation in hepatic blood flow delay biotransformation
 - Enzyme inhibition leads to toxicity
 - Two phases of biotransformation
 - Phase I: oxidation, reduction, hydrolysis
 - Phase II: conjugation

Toxicokinetics (cont.)

• Excretion

- Final means of elimination as metabolites or unchanged
- Routes include
 - Kidneys: water soluble toxicants
 - Lungs: volatile substances
 - Bile: those undergoing enterohepatic circulation
- **Time of elimination = $V_d / \text{clearance}$**
 - Determines period required to remove the toxicant by dialysis
 - Toxicants with larger V_d take longer time of dialysis
 - Used to know the contribution of each organ to total clearance to look for strategy for elimination of toxicants

Toxicokinetics (cont.): question 2

- For toxicants 95% cleared by liver metabolism and only 5% cleared by renal excretion, would dramatic diuresis affect overall elimination much? explain

Toxicokinetics (cont.)

- Drug over- dosage can alter the usual kinetics value
 - Injure GIT leading to altered absorption
 - Exceed the capacity of liver to metabolize
 - Plasma protein binding could be exceeded
 - At drugs follow zero order kinetics of elimination

Toxicodynamics

Mechanisms of Action

- ADEs can occur at the level of the molecule, cell, organ, or organism
- How important
 - Rational basis for interpreting descriptive toxicity data
 - Estimating the probability of harmful effect
 - Provide preventive procedures or antagonize toxic effects
 - Designing drugs and industrial chemicals that are less hazardous
 - Develop selective pesticides
 - Understand fundamental physiology processes ranging from Neurotransmission (curare), through DNA (alkylating agents)

Mechanisms of Action

- **Molecularly:** chemical can interact with
 - Proteins, Lipids, DNA
- **Cellularly:**
 - receptor-ligand binding
 - membrane function
 - cellular energy production
 - bind to biomolecules
 - perturb homeostasis (Ca)

Effects of Toxicants on Target Molecules

- Ultimate result of toxicant interaction with endogenous molecules may cause
 - **dysfunction or**
 - **distraction of target molecules.**
- Some mimic endogenous Ligands
 - **Morphine activates opioid receptors**

MOT ...

- **Ion channels**
 - Atropine, curare and strychnine (Block NT receptors & interfere ion channels)
- **Mitochondrial electron transport complexes**
- **Enzymes** (irr AchEI)

Two types of dose-response relationships

- Individual dose response curve
 - Response of an individual to varying doses of a chemical or drug
 - Graded response – continuous over a range of doses
- Quantal dose response curve
 - Population
 - Distribution of responses to different doses in a population of individual organisms

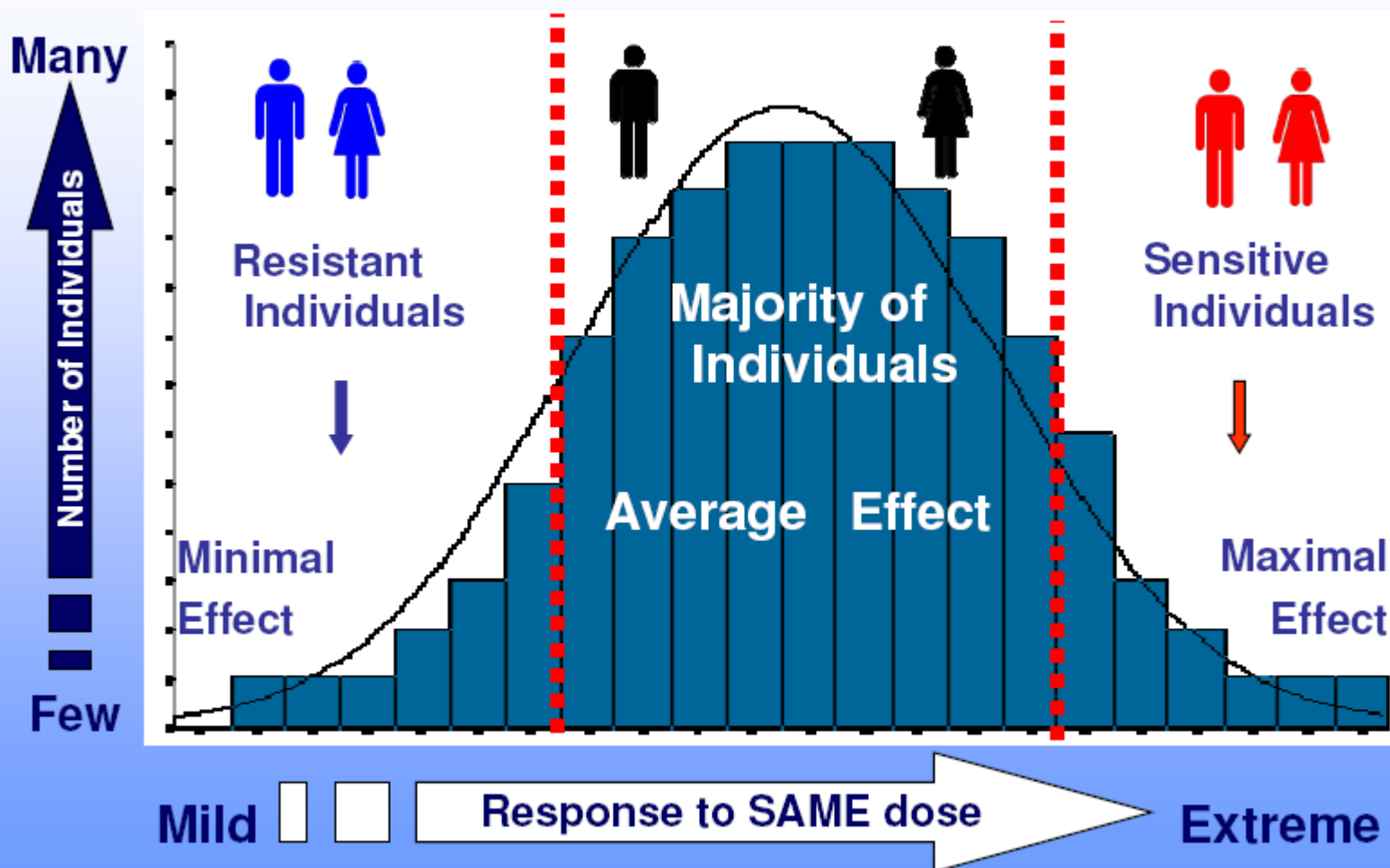
Quantal Dose-Response

- All or none
- Population
- At a given dose, an individual in the population is classified as “responder” or “non-responder”
- Responses are conceptually identical

All or None or Graded Response

- All or None – Live or Dead
 - Percentage of the animals or cells following dosage show a response
- Graded Response – Pathological Damage
 - Actual values plotted against dosage

Population Dose-Response



© 2008 Society of Toxicology

Parameters Derived from the Curve

- When lethality has been used as an endpoint, the LD_{50} can be determined.
 - LD_{50} is defined as the dosage of a substance which kills 50 percent of the animals in a particular group (acute dose)
 - Varies by different groups and species
 - Give information on how toxic the compound is to other substances.

LD 50

- An LD50 represents the individual dose required to kill 50 percent of a population of test animals. It is an index determination of medicine and poison's virulence. The lower the LD50 dose, the more toxic the pesticide.

LD_{50} : Median lethal dose

- LD_{50} (median lethal dose):
 - The dose of a drug that produces death in 50% of the animal population tested

Drug variability and toxicity assessment

- ED₅₀: Effective dose for 50% of subjects
- LD₅₀: Lethal dose for 50% of subjects
- The therapeutic index

$$\mathbf{TI} = \mathbf{LD_{50} / ED_{50}}$$

No drug is 100% safe

LD 50s

- Ethanol - 10,000 mg/kg
- Nicotine – 1 mg/kg
- Botulinus toxin – 0.00001 mg/kg
- Threshold dose
 - Minimal effective/toxic dose of any chemical or drug

PRINCIPLES OF TOXICOLOGY

Table 2-2. TOXICITY RATING CHART

TOXICITY RATING OR CLASS	PROBABLE LETHAL ORAL DOSE FOR HUMANS	
	<i>Dosage</i>	<i>For Average Adult</i>
1. Practically nontoxic	> 15 g/kg	More than 1 quart
2. Slightly toxic	5-15 g/kg	Between pint and quart
3. Moderately toxic	0.5-5 g/kg	Between ounce and pint
4. Very toxic	50-500 mg/kg	Between teaspoonful and ounce
5. Extremely toxic	5-50 mg/kg	Between 7 drops and teaspoonful
6. Supertoxic	< 5 mg/kg	A taste (less than 7 drops)

CASARETT AND DOULL'S, TOXICOLOGY, THIRD EDITION

Table 2-1. APPROXIMATE ACUTE LD50'S OF SOME REPRESENTATIVE CHEMICAL AGENTS

AGENT	LD50 (mg/kg)*
Ethyl alcohol	10,000
Sodium chloride	4,000
Ferrous sulfate	1,500
Morphine sulfate	900
Phenobarbital sodium	150
Picrotoxin	5
Strychnine sulfate	2
Nicotine	1
<i>d</i> -Tubocurarine	0.5
Hemicholinium-3	0.2
Tetrodotoxin	0.10
Dioxin (TCDD)	0.001
Botulinum toxin	0.00001

* LD50 is the dosage (mg/kg body weight) causing death in 50 percent of the exposed animals.

From Casarett & Doull's, Toxicology
3rd Edition, 1986

Other terms

- Similar to LD_{50}
- ED_{50}
 - Effective dosage for 50 percent
- TD_{50}
 - Toxic dosage for 50 percent
- Therapeutic Index
 - Ratio of LD_{50} or TD_{50} to ED_{50}
 - LD_{50}/ED_{50} or TD_{50}/ED_{50}

The larger the value the greater is the margin of safety between the effective dose and toxic dose.

Factors Influencing Toxicity

- form and innate chemical activity
- dosage, especially dose-time relationship
- exposure route
- Species
- Age
- Sex
- ability to be absorbed
- Metabolism
- distribution within the body
- Excretion
- presence of other chemicals

Characteristics of exposure

- Toxic agent or metabolic break-down (biotransformation) product must reach its target site in the body
 - at a concentration and for a length of time sufficient to produce a toxic manifestation.
- Whether a toxic response occurs is dependent on
 - the chemical and physical properties of the agent,
 - the exposure situation,
 - how the agent is metabolized by the system, and
 - the overall susceptibility of the biological system or subject.

Route and Site of Exposure

- Gastrointestinal tract
 - ingestion
- Lung
 - inhalation
- Skin
 - Topical, dermal

Greatest effects

- Intravenous route
- Inhalation
- Intraperitoneal
- Subcutaneous
- Intramuscular
- Oral
- Dermal
- Vehicle – material chemical is dissolved

Duration and Frequency

- Acute
 - Less than 24h, single incident
- Subacute
 - 1 month or less, several weeks of exposure
- Chronic
 - More than 3 month (over months or years)

Human Exposure Situations

- Frequency and duration are usually not clear
- Elimination rate is important to toxicity

Frequency Of Exposure

Number of Times

Time Between

4 Beers in an hour or

4 Beers in 4 Days

Spectrum of undesired Effects

- Broad
- Drug = one desirable effect
- Other effects are undesirable or side-effects
- Side effects can be therapeutic (another condition), adverse, deleterious, or toxic

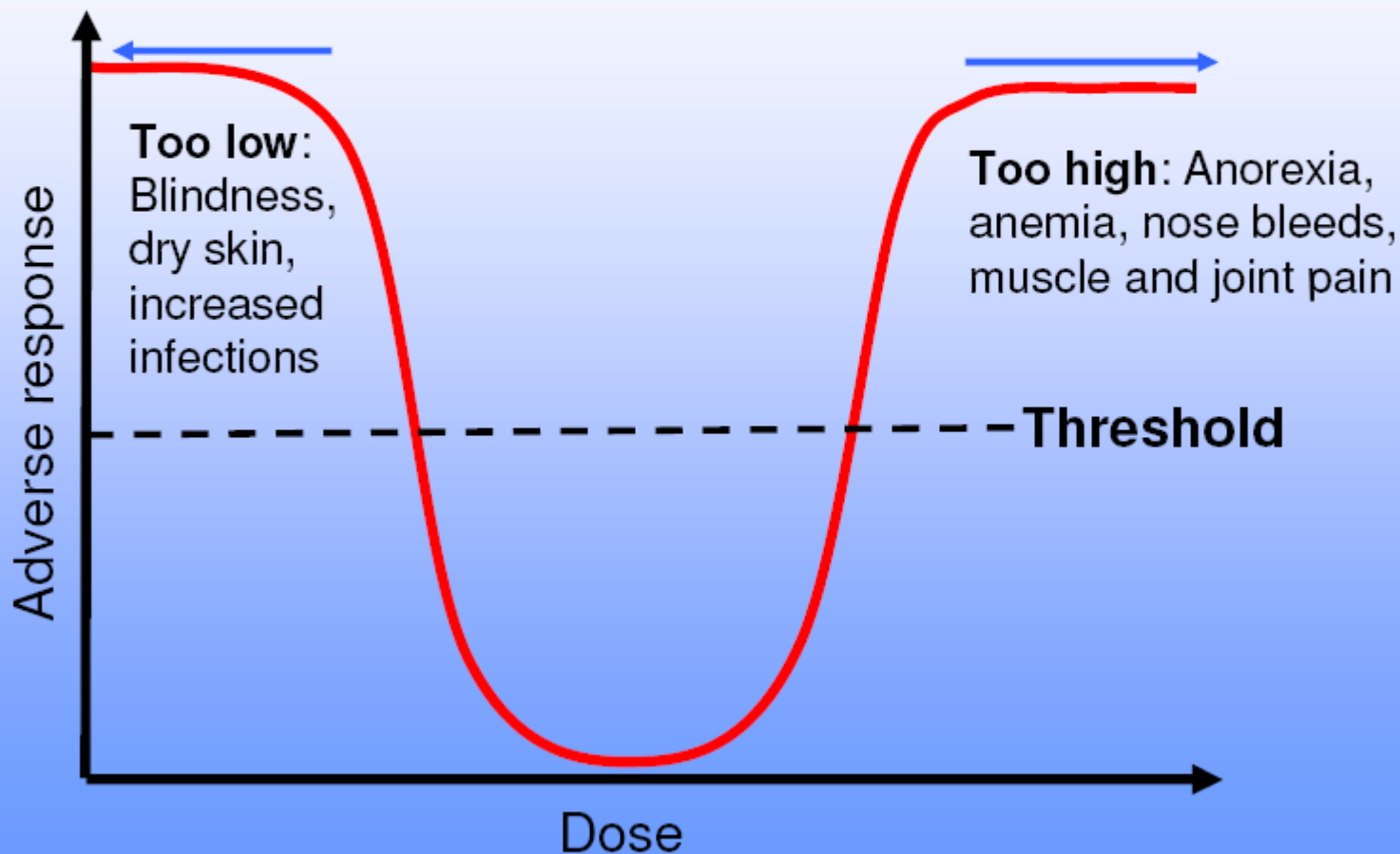
Route of Administration and LD₅₀

- Pentobarbital
 - Oral - 280 mg/kg
 - Subcutaneous – 130 mg/kg
 - Intramuscular – 124 mg/kg
 - Intravenous – 80 mg/kg

Species Differences and LD₅₀

- Rabbit – 40 mg/kg Lung damage
- Mouse – 20 mg/kg kidney and lung damage
- Rat – 12 mg/kg Lung Damage

Some chemicals have both therapeutic and toxic effects: Vitamin A



Animals

Rodent and non-rodent

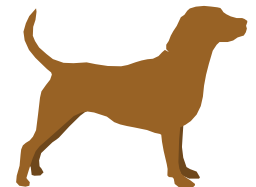
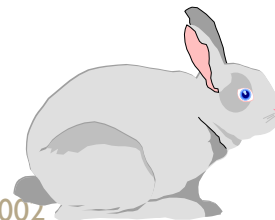
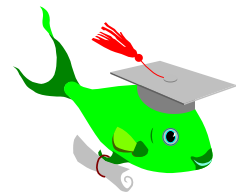
Fish

Aquatic invertebrate

non primates

dog

rabbit



Categories of toxicological tests

- Acute toxicity testing: single dosage everyday, to observe its toxic reaction after 7 days and 14 days.

Toxicity Testing

- Acute toxicity
- Subacute toxicity
- Chronic toxicity
- Reproductive toxicity
- Genotoxicity
- Neurotoxicity
- Immunotoxicity

- 
- Acute toxicity test: that is lethal in approximately 50% of animals. Determine maximum tolerated doses. Usually two species, two routes, single dose (compare with therapeutic dose).

Spectrum of Toxic Dose

- A poison is any substance by any route at any time which causes a deleterious response or effect in an organism

Systemic toxic effects

- Categorized according to the site of the toxic effect.
- In some cases, the effect may occur at only one site.
 - **specific target organ.**
- In other cases, toxic effects may occur at multiple sites.
 - **systemic toxicity.**
- Following are types of systemic toxicity:

Acute Toxicity

- Occurs almost immediately (*hours/days*) after an exposure.
- An **acute exposure** is usually a single dose or a series of doses received within a 24 hour period.
- Death is a major concern in cases of acute exposures.
 - Examples are:
In 1989, 5,000 people died and 30,000 were permanently disabled due to exposure to methyl isocyanate from an industrial accident in India.
 - Many people die each year from inhaling carbon monoxide from faulty heaters.
- Non-lethal acute effects may also occur, e.g., convulsions and respiratory irritation.

Acute Lethality



Acute Lethality

- First test performed
- LD50
 - Several routes of administration
 - One or more species
 - Mouse, Rat, Rabbit, and dog
 - Daily examination of animal
 - How many die after a single dose? (14 days)

Acute Lethality

- Data generated
 - Quantitative estimate of toxicity
 - Identify target organs
 - Reversibility of toxic response
 - Information of dose

Skin and Eye Irritation

- **Dermal irritation test**
 - Draize test
 - Skin of rabbit is shaved and chemical applied to one area and covered for 4 hours
 - Degree of irritation determined (red, edema, etc)
- **Ocular irritation**
 - Chemical instilled into one eye of the test rabbit

Sensitization

- For chemicals that come in contact with the skin
- Chemical is administered to the shaved skin of guinea pigs
- Observe for 2-4 weeks
- Look for erythema (redness)

Subchronic toxicity

- Results from repeated exposure for several weeks or months.
- This is a common human exposure pattern for some pharmaceuticals and environmental agents.
 - Examples are: Ingestion of coumadin tablets (*blood thinners*) for several weeks as a treatment for venous thrombosis can cause internal bleeding.
 - Workplace exposure to lead over a period of several weeks can result in anemia.
- Last for 90 days
- LOAEL (lowest observed adverse effect level)
- NOAEL (no observed adverse effect level)

Chronic toxicity

- *Chronic toxicity* represents cumulative damage to specific organ systems and takes many months or years to become a recognizable clinical disease.
- Damage due to subclinical individual exposures may go unnoticed.
- With repeated exposures or long-term continual exposure, the damage from these subclinical exposures slowly builds-up (*cumulative damage*) until the damage exceeds the threshold for chronic toxicity.
- Ultimately, the damage becomes so severe that the organ can no longer function normally and a variety of chronic toxic effects may result.

- 
- Chronic toxicity test: 1 to 2 years.
Required when drug is intended to be used in humans for prolonged periods.
Usually run concurrently with clinical trial.
 - Goals of chronic tests are to show which organs are susceptible to drug toxicity.
Clinical chemistry physiologic signs, autopsy studies, hematology, histology, electron microscopy studies. Identify target organ of toxicity.

Chronic toxicity contd.

Examples of chronic toxic affects are:

- cirrhosis in alcoholics who have ingested ethanol for several years
- chronic kidney disease in workmen with several years exposure to lead
- chronic bronchitis in long-term cigarette smokers
- pulmonary fibrosis in coal miners (*black lung disease*)

Carcinogenicity

- is a complex multistage process of abnormal cell growth and differentiation which can lead to cancer.
- At least two stages are recognized.
 - **initiation** in which a normal cell undergoes irreversible changes and
 - **promotion** in which initiated cells are stimulated to progress to cancer.
- Chemicals can act as **initiators** or **promoters**.

Developmental Toxicity

- pertains to adverse toxic effects to the developing embryo or fetus.
- This can result from toxicant exposure to either parent before conception or to the mother and her developing embryo-fetus.
- The three basic types of developmental toxicity are:
- Embryoletality- failure to conceive, spontaneous abortion or stillbirth
- Embryotoxicity- growth retardation or delayed growth of specific organ systems
- Teratogenicity- irreversible conditions that leave permanent birth defects in live offspring (eg. Cleft palate, missing limbs)

Developmental Toxicity contd.

- Chemicals cause developmental toxicity by two methods.
- They can act directly on cells of the embryo causing cell death or cell damage, leading to abnormal organ development.
- might also induce a mutation in a parent's germ cell which is transmitted to the fertilized ovum. Some mutated fertilized ova develop into abnormal embryos

Genetic Toxicity

- results from damage to DNA and altered genetic expression.
- This process is known as **mutagenesis**.
- The genetic change is referred to as a **mutation** and the agent causing the change as a **mutagen**.
- There are three types of genetic change:
 - Gene mutation-change in DNA sequence within a gene
 - Chromosome aberration-changes in chromosome structure
 - Aneuploidy/polyploidy-increase or decrease in no of chromosomes